# Mahalingam Ramasamy

Software Engineer

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## Summary

Software Engineer with 2+ years owning production systems end to end in Node.js (NestJS), Java (Spring Boot) and React, from API design to AWS deployment. Delivered event-driven services, real-time WebSocket APIs, payment integrations and a multi-provider LLM layer with automatic failover.

## Experience

**Ekko Medical Solutions Pvt Ltd** **Oct 2025 – Present**
*Software Developer (Backend)* *Chennai, India*

- Building an **AI medical scribe** that turns doctor–patient conversations into clinical notes; cut note **latency by 50% (50s to 25s)** with a lightweight refinement LLM and event-driven async generation.
- Architected a provider-agnostic LLM layer across **OpenAI, Gemini and Anthropic** powering **15+ AI tasks**, routing each task to the best-fit model with same-request failover to a backup provider.
- Engineered a horizontally scalable real-time transcription microservice (**WebSockets, Redis, PostgreSQL**) delivering sub-second live transcripts for sessions up to **5 hours**, with automatic recovery on disconnect.
- Built **Cashfree** subscription billing with **UPI/eNACH autopay across 7 plans**, HMAC-verified webhooks, idempotent processing and hourly reconciliation.

**ROX Hi-TECH** **Jun 2024 – Oct 2025**
*Integration Engineer (Full Stack)* *Chennai, India*

- Developed a full-stack HR platform (**Java, Spring Boot, React**) for **TNEB**, Tamil Nadu's state electricity board, digitizing leave, transfer and performance-review requests for **100,000+ employees**.
- Replaced paper-based approvals with a configurable approval engine across **6+ request types**, cutting turnaround by **70%** and letting new workflows launch through configuration instead of code changes.
- Built **Spring Boot REST APIs** integrating **SAP ERP** via **OData** to sync employee master data, eliminating duplicate data entry between the HR platform and SAP.
- Implemented **role-based access control** across **3 user roles** with **Spring Security**, enforcing approvals at the API layer so each role can view and act only on requests within its scope.

## Projects

**Shoppy – E-Commerce Platform** | *Java, Spring Boot, Next.js, PostgreSQL, Stripe, AWS* **GitHub**

- Integrated **Stripe** payments with signature-verified, **idempotent webhooks**, guaranteeing exactly-once order creation even when Stripe retries events.
- Secured APIs with **Spring Security, JWT** and role-based access, and pushed live product updates to the Next.js storefront over **WebSockets (STOMP)**.
- Wrote **54 automated tests** (JUnit 5, Mockito, Testcontainers) reaching **80% coverage**, run in CI on every push.
- Automated deployments to **AWS** with CodePipeline and **Elastic Beanstalk**, serving data from RDS PostgreSQL and assets from S3 behind CloudFront.

## Technical Skills

**Languages**: Java, TypeScript, JavaScript, Python, SQL
**Backend**: Node.js, NestJS, Spring Boot, FastAPI, REST APIs, WebSockets, SSE, Celery
**Databases**: PostgreSQL, pgvector, Redis, Prisma
**Cloud**: AWS (ECS Fargate, ALB, Lambda, S3, CloudFront, SES, VPC, Secrets Manager), GCP
**DevOps**: Terraform, Docker, GitHub Actions, CI/CD
**Frontend**: React, Next.js, Tailwind CSS

## Education

**SASTRA Deemed University** **Jun 2020 – Jun 2024**
*Bachelor of Technology in Computer Science and Business Systems; CGPA: 7.2/10* *Thanjavur, India*